Computer Games Development CW208
GDD
Year IV

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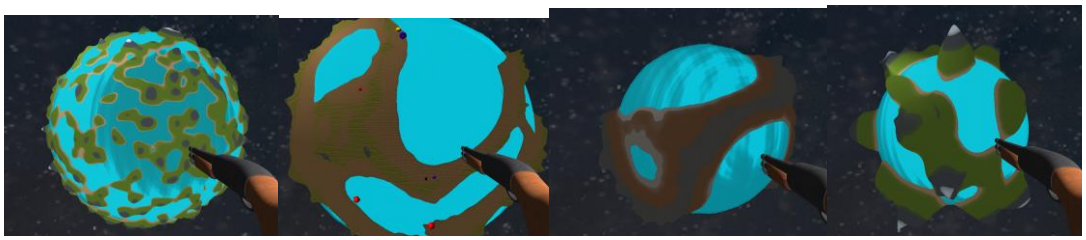
Dr. Noel O'Hara - who gave many insightful ideas and thought on the project as a whole

All the supervisors who gave me some good ideas on what to do with the gameplay, and what direction I should take with the planets

Game Overview

My game is a 3D open world procedurally generated planet explorer.
The game allows for the interaction with unique planets

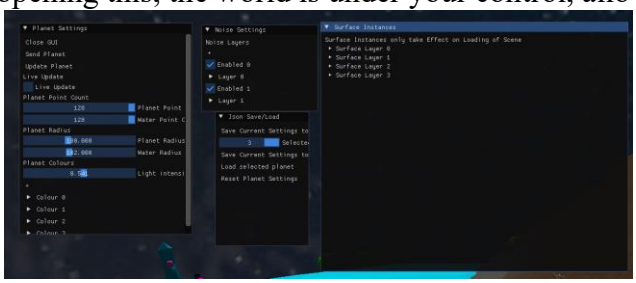
Here are screenshots of just some of the generated planets:



You travel through the planets fighting different enemies and exploring the unique planets until you die from the enemies.

The 2 enemies who shoot at you or melee you close range are the shard and slime respectively.

There is an in built level editor which can be accesed by typing /gui in the chat window. After opening this, the world is under your control, allowing you to change it to whatever you feel.



Common Questions:

What is the game?

This is a planet exploring game where you can explore procedurally generated planets

Why create this game?

This game is a demonstration of what kind of interesting explorable terrain can be created in OpenGL

Where does the game take place?

In space, on the procedurally generated planet

What do I control?

You control the player who is exploring the planet and attempting to survive

How many characters do I control?

1

What is the main focus?

Explore the unique planets available to you

What's different?

The planets change on every run, making every time you load up a unique experience

Feature Set

General Features

Procedurally generated planets allowing for interesting and unique gameplay every time you load in

Editor

Comes with a world editor that is useable in game, allows for the save and load of new and unique planets

Gameplay

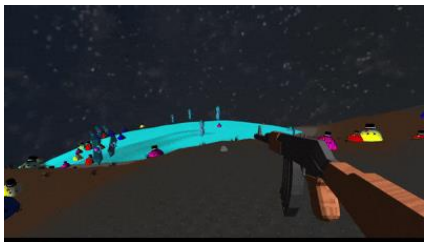
Different weapons
interesting terrain to walk around
different enemy types with interesting attack styles

The Game World

Overview

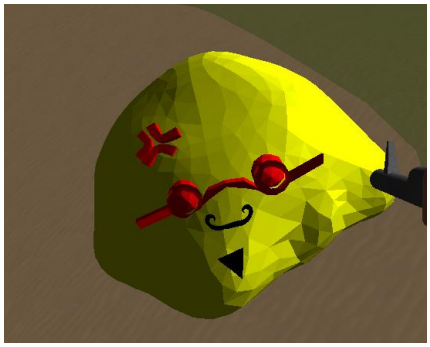
Every world is unique

The game world is unique every time you load up, maybe the islands are different, maybe there's very few, maybe there's many. The terrain could be mountainous or flatter, who know until you try it out.



Planet specific enemies

Each planet will have a unique twist to the enemies, maybe they'll be wearing sunglasses or maybe they are fancy and wear top hats.



The Physical World

Overview

The world is unique on every play through, so youll never know what you get. It could be chaotic, or it could be peaceful. Play a little and find out.

Key Locations

The entire world is key, and unique

Travel

Classic WASD to move forward back left and right. Moving around in first person!

Scale

The scale of the world depends on your luck, it could be a tiny moon you end up on, or a huge water planet.

Objects

Slimes are the main inhabitants of these worlds!

Time

The player will walk around the world for the first 30 seconds then will encounter an enemy, and kill it, in the next 30 seconds they will encounter more enemies and try to kill most of them, by 1.30 they will be dead. Then they will restart and try again.

Rendering System

Overview

The game will use a complex graphics pipeline with shaders, and 3D models. They will all be called through a standard draw call that will happen after every update.

2D/3D Rendering

3D using OpenGL

Camera

Overview

The camera will move around with your player in a first-person perspective.

Camera Detail #1

First person movement, using mouse

Game Engine

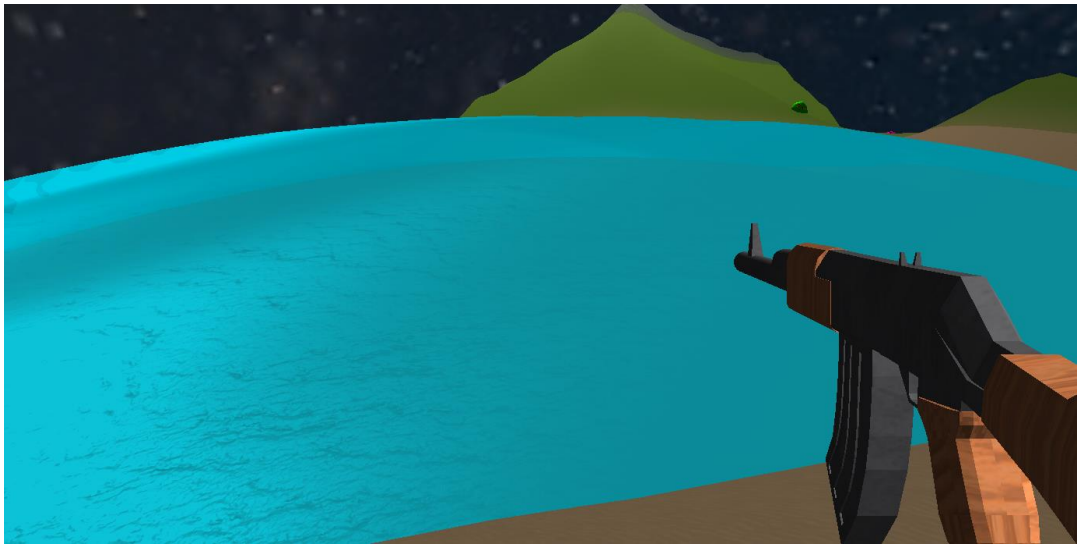
Overview

No Game engine, built from ground up to mimic a gane engine

The game engine uses a game object approach, where every game object has a list of scripts and tags. The scripts will be an abstract class that allows for any unique behaviour that is intended to be inserted into the script. All actions will be decoupled and allow for any function that is needed to be called, while avoiding unnecessary ones.

Water

The water will be an ICO sphere that is deformed by shaders and coloured by them.



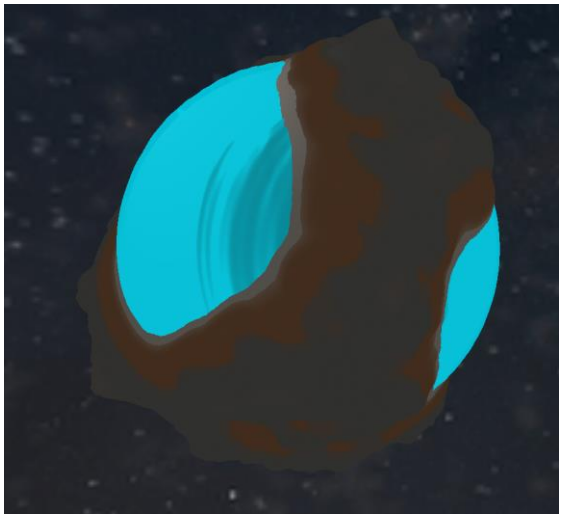
Collision Detection

The game only has simple floor collision which uses the player position which samples the noise map providing the players elevation.

The World Layout

Overview

The world is a sphere that you can walk around, the world is randomly generated and may look like this:



Or this

Who knows, until you try it out.

Game Characters

Overview

You the player are the character, you are trying to survive on this alien planet.

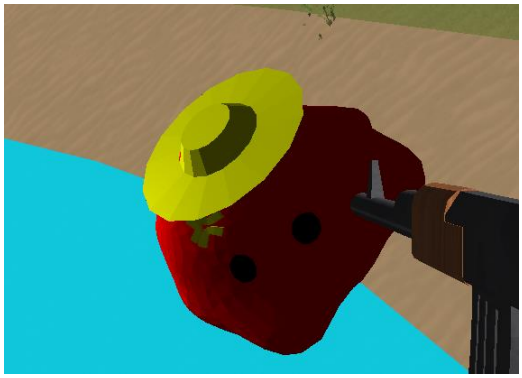
Enemies and Monsters

There are 2 unique enemies

The slime

The slime comes in many colours and sizes, the slime also comes with many hats and glasses, depending on what planet you are on (different enemies appear on different planets).

The slime wanders around and idles around a bit until it detects you
when it detects you it moves in range to attack, but before it attacks it charges up its dash to damage you, once it is charged upit turn red and runs towards you very quickly. If you dont avoid it you take damage.
After its dash it takes 3 seconds to reset, and then will try to dash at you again, it will keep doing this until you die, or it dies.



The Shard
Usually found at the tops of mountains; the shard will remain stationary waiting for you to approach it. Once you are in range it will start shooting at you, when it turns red you must dodge its bullets. The shard shoots in 3 round bursts until it must cool down. I will not seek you out and much prefers the comfort of mountain ranges.

User Interface

Overview

The user interface for the main menu is very simple, play or quit. When you play you have no user interface, but you do have the option to open the chat window. When the chat window is opened (using the enter key) you can write different commands.

GUI

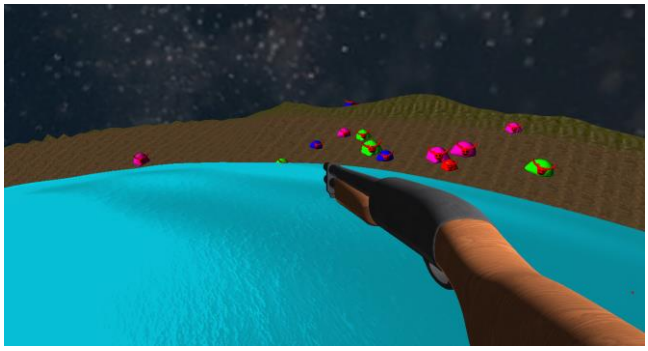
The main command is /gui , which allows you to edit the world through a new user interface window. It allows for the manipulation of the world using the different buttons and sliders.

Weapons

Overview

You have the direct use of 3 simple weapons. (Middle mouse button allows for the switching of weapons)

The shotgun

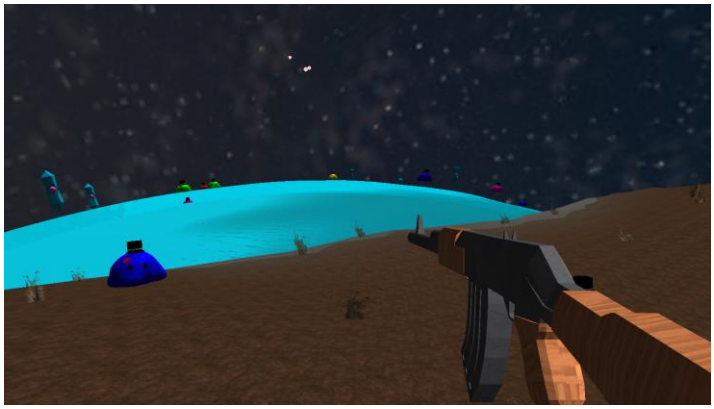


High damage, long reload, high recoil, low fire rate

The Pistol

As effective as you make it
Shoots as fast as you can press the mouse
low damage, fast reload, low recoil, very fast reload

The rifle



Full auto, medium damage, moderate fire rate, low recoil fast reloads

Single-Player Game

Overview

The player can walk around and explore the world as much as they want, hopefully finding and killing as many enemies as they can

Hours of Gameplay

The gameplay is roughly 10 –15 minutes, surviving on each world variant a few minutes.

Victory Conditions

You cant win, there is no ending.

Persistence

No persistence is saved during reloads

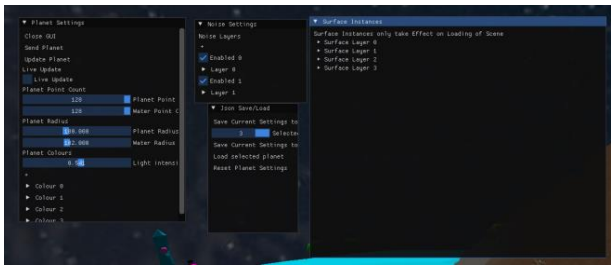
Saving and Loading

The worlds you create are saved / loaded

World Editing

Overview

The world editor is the central element to the experience in this game. It allows for the creation of unique worlds with as much variance as you can think of.



Update Planet

This updates the current planet to use the newest settings

Live Update

This makes it so that every change made is applied to the planet instantly (very laggy)

Planet point Count

This controls the amount of vertices the planet and water will have

Planet Radius

Changes the size of the planet

Light intensity

How bright the planet textures are

Planet Colours

Each planet layer colour has controls of texture or colour, if a texture is applied a normal map can also be used, you can control the tiling density and normal height, you can also control at what layer the texture appears at

Noise Layers

The noise layers settings allow for controls over how the planet is deformed, this is through 2 different noise functions, either normal (which applies a normal type of noise which produced smooth bumps) and rigid, which provides more spiky terrain which is suitable for mountains. The other settings adjust the layer height, repititions, scale and miniumum height values, all providing fine controls over the noise settings

Json Save / load

The Json save / load function helps the player save and load planets from / to a json folder. Gives access like save to current, save to new load and reset.

Surface Instances

Surface instances allow for control of what is spawned on each layer of the world. You can instantiate grass, models, pickup models, and enemies. Each instance type also has different controls that change how the instances are spread around the map and how dense they spawn. Certain variants (like enemies) have even more fine tuned controls, these dont have to be used but can speed up the placing of enemies. For instance, all the slimes have unique texture variants, so I use the model duplicates settings to make it fast to add more variants.